# Energy-Aware Arm and Base Selection for a Dual-Gantry Quad-Arm Ceiling Robot Using a GNG Capability Map

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***Abstract*—The two ceiling-mounted rails carrying four arms turn every pick into two coupled decisions, which arm should act and where its base should go, since repositioning the base is a dominant cost of the motion. We propose an energy-aware selection cost $J$ that chooses the arm and the full eight-degree-of-freedom (8-DOF) configuration together over a base-aware capability map built with Growing Neural Gas (GNG). Every node stores a representative configuration, rail placement included, and its manipulability, so it already fixes both an arm and a base rather than marking only whether a cell is reachable. Treating the rail as a degree of freedom when building the map enlarges the measured reachable workspace by about four times, and boundary-seeding keeps the map accurate to within about a centimetre of the true reach surface. In simulation on NVIDIA Isaac Sim, energy-aware selection is evaluated against nearest, fixed, and random baselines under a rail dynamics model with assumed Coulomb/viscous friction (a modelling assumption, not measured hardware data), so repositioning the rail carries a real energy cost. Over a 60-position grid, the multi-arm policies each reach 93.3% plan success versus 63–65% for any single fixed arm, and energy-aware selection attains the lowest mean measured mechanical energy of the three (5.1% below nearest, 3.1% below random), with a majority (57–61%) same-target pairwise win rate against each baseline.**

***Index Terms*—mobile manipulation, capability map, growing neural gas, redundant manipulator, base placement, energy-aware planning**

## I. Introduction

An aging global population is driving demand for assistive robots that can operate inside the home [1], taking on tasks such as retrieving objects, tidying, and delivering items to occupants [2]. Residential deployment is more challenging than factory or outdoor use [3], because homes are small and cluttered, and a mobile manipulator shares that space with residents who must stay alert to avoid it. Lifting the robot off the floor entirely [4] removes this conflict. A ceiling-mounted arm keeps the floor unobstructed while still serving as an in-home assistant from above [5]. We study such a ceiling robot with a dual-gantry system in which each gantry carries two 6-DOF arms, forming a quad-arm cooperative overhead manipulator.

Each gantry moves both linearly and rotationally, giving each arm 8 DOF in total. Most objects are reachable by several arms from a range of base positions, so the robot faces two coupled questions, which arm should act and where should its base be positioned first? Both are costly because repositioning the heavy base costs time and energy [6]–[8] and naive policies (nearest arm, or leaving the base where it is) can waste base travel. We address the problem of selecting the arm and base placement that minimizes energy.

Reachability and capability maps are the standard tool for depicting an arm's reachable regions and mobility, and for placing a robot optimally [9], [10], but two assumptions limit them here. Most are built for a fixed base, so they miss the reachable space a mobile base opens up. Also, they typically record only whether a cell is reachable, not the cost of reaching it or an IK-ready configuration [9]. Recent work improves operator awareness of obstacles and nearby objects [6], and redundancy-resolution methods optimize a single arm's posture for manipulability [11], but neither selects between multiple arms nor accounts for the base's own cost. What is missing is a representation that answers, directly, which arm and which base placement minimize energy on a ceiling-rail robot.

We address this with an energy cost $J$ that operates on a base-aware capability map built with Growing Neural Gas (GNG) [12]. Each node in this GNG map stores a representative 8-DOF configuration and its manipulability [11], so a node is a full configuration of base and arm not just a reachable point. This paper makes three primary contributions:

- **An energy-aware selection cost** $J$ that optimizes the choice of arm and the complete 8-DOF configuration (including rail travel), rather than relying on the nearest reachable pose.
- **A base-aware GNG capability map** that treats the ceiling rail as an active DOF and associates a representative configuration and manipulability value with every node, using a boundary-seeding technique to preserve accuracy at the workspace edge.
- **A comprehensive evaluation in Isaac Sim** that characterizes the map, quantifies the workspace expansion enabled by the rail DOF, and compares our energy-aware selection policy against nearest, fixed, and random baselines.

Additionally, an open-vocabulary perception pipeline utilizing YOLOE provides the target objects for the selection framework [13].

Our simulation results validate these claims. Treating the rail as an active DOF expands the reachable workspace by approximately $4\times$ compared to an arm-only baseline, while boundary-seeding recovers the true reach at the workspace edge to within about one centimeter of the forward-kinematics surface. Energy-aware selection, alongside the other multi-arm baselines, increases planning success from 63–65% (for any single fixed arm) to 93.3%. Under a rail-dynamics model with assumed Coulomb and viscous friction, so that rail travel carries a realistic energy cost, our policy attains the lowest mean mechanical energy among the three multi-arm strategies (45.84 J vs. 48.28 J for the nearest baseline [$-5.1\%$], and 47.31 J for the random baseline [$-3.1\%$]). This yields a majority same-target pairwise win rate of 57–61% against each baseline, although the median performance remains closer to the random policy. We ground these results by utilizing NVIDIA Isaac Sim [14] as a high-fidelity digital twin of our physical trailer living laboratory and ceiling-mounted robot system [15]. While physical integration and real-world validation are reserved for future work, both the trailer lab and the ceiling hardware already exist in our facility, demonstrating the feasibility of transferring the simulated controller.

## II. Related Work

**Capability maps and base placement.** Zacharias et al. extended binary reachability grids to a capability map that also captures reach orientation [9], and Reuleaux inverted this to pick a standing position for a target [10]. MoMa-Pos optimizes base placement from object kinematics [6], and BaSeNet plans a sequence of base poses for successive pick-ups [7]. This body of work places a *single* arm well from a stationary-base map storing one reachability bit per cell, but none chooses *among* several arms or scores placements by energy. That is the gap our work fills. Our map differs on both counts, since the ceiling rail is a first-class part of the configuration, so it records how base motion extends reach, and every node carries a representative 8-DOF configuration and manipulability value, not a single bit.

**Redundancy resolution and inverse kinematics.** A kinematic chain with more joints than the task strictly needs can use the spare freedom for a secondary objective; Yoshikawa's manipulability measure is the classic choice, steering a redundant arm away from singular postures [11]. Such methods optimize the posture of a single, already-chosen arm for one criterion, and none decides which of several limbs should act, or accounts for the base travel a given posture would require. We keep the IK solver standard and classical, and instead use the capability map to enumerate candidate 8-DOF configurations, each with its own arm and rail placement, which the energy cost then ranks to settle both the arm choice and the base motion.

**Multi-arm coordination.** Distributing tasks across multiple arms is a related concern; recent planners such as RoboPARA build dependency graphs over tasks and allocate subtasks to each arm so multiple arms can work in parallel [16]. Their focus is the temporal and logical structure of a multi-step task. Ours is different and complementary. For a single reaching action, we ask which arm reaches the target for the least energy, since arms sharing a rail share one moving base while arms on different rails move independently, and we answer it with the same map and cost that also place the base.

## III. System

### A. Robot Platform and Kinematic Model

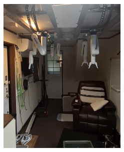
The robot has two ceiling rails side by side, each carrying two Kinova Gen3 Lite arms (six joints, $\sim$0.76 m reach, two-finger gripper [17]; four arms total). Each rail adds two DOF its pair of arms share, a prismatic axis sliding the carriage horizontally and a revolute axis rotating it, so relative to the ground each arm forms an 8-DOF chain, written $\mathbf{q} = [d,\ \theta,\ q_1,\ \ldots,\ q_6]$ ($d, \theta$: that arm's own rail translation/rotation; $q_1 \ldots q_6$: arm joints). The kinematic tree is anchored at a fixed world frame, with each rail base attached to the world, each arm base to its own rail, and a tool frame at the gripper defining the end-effector pose. Since the two arms on one carriage move together, moving the rail couples the arm choice to the base placement, while arms on different rails move independently. Fig. 1 shows the platform, its frames, and the end-to-end pipeline from perception through selection to planning and execution.

### B. Simulation Environment

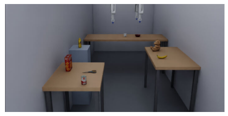
All experiments in this paper are carried out in NVIDIA Isaac Sim, a GPU-accelerated simulator with a physics engine suitable for contact-rich manipulation [14]. The simulated robot is built from the same URDF description that defines the physical platform, so the kinematics and joint limits used in simulation match the real hardware. The scene places a set of everyday tabletop objects within the workspace, some within easy reach of several arms and at least one deliberately positioned near the edge of the reachable workspace (Fig. 2). The scene is observed by two RGB-D cameras mounted near the ceiling at diagonally opposite corners of the room, each aimed down and across the workspace toward the tables under the arms; the two viewpoints overlap in the middle of the pick area, so an object occluded from one camera is usually still visible from the other (Fig. 2 shows the resulting views). These cameras are the system's only exteroceptive sensor, and their images serve two roles at once. The color stream feeds the object detector, and the depth stream builds the collision octomap used for planning, so a single sensing rig supports both perception and collision avoidance. We treat the simulator as the evaluation platform for this work and validate on the physical robot in future work.

### C. Perception Front End

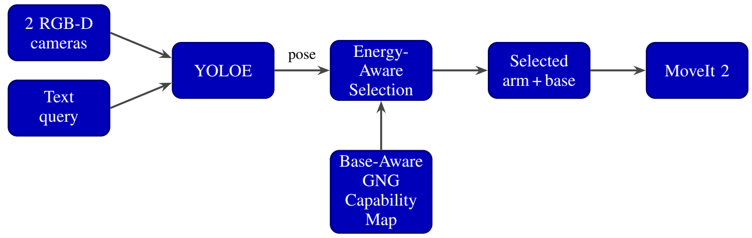
The target objects are supplied by an open-vocabulary perception module based on YOLOE [13]. Each overhead camera is processed independently, so a per-object mask's depth pixels are back-projected to a 3D point set in the world frame, and we take the *median* of those points as the object's centroid (robust to mask-border depth outliers). Detections



(a) physical platform



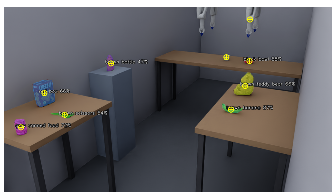
(b) Isaac Sim twin



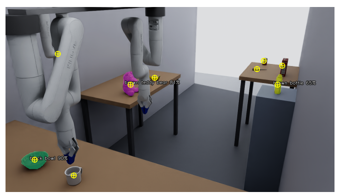

(c) end-to-end pipeline

Fig. 1. Overview of the robot and pipeline. (a) Physical platform in our trailer living laboratory (*all experiments in this paper are in simulation*). (b) Isaac Sim digital twin. (c) End-to-end pipeline.

from the two cameras are fused by centroid proximity (0.10 m merge radius, positions averaged, higher observed top kept) and stabilised by a track layer with majority label voting, giving a single world-frame position per named object without a fixed object database (Fig. 2). Perception is the enabling front end that supplies the target for the capability map and energy cost.



(a) camera 1



(b) camera 2

Fig. 2. Open-vocabulary perception, viewed from (a) camera 1 and (b) camera 2.

### *D. Planning and Execution Stack*

Given a target and a candidate configuration, the motion layer is built on MoveIt 2 [18] with planners from the Open Motion Planning Library [19]. Inverse kinematics requests are answered by MoveIt against the current robot state, and collisions are checked against an octomap populated from the two cameras' depth images, so a plan avoids both the scene and the robot's own body; the object selected for removal is isolated from the octomap's source cloud so it can be grasped, while the remaining objects and the environment stay as obstacles. We build the capability map of Section IV offline with the Pinocchio rigid-body-dynamics library [20], which is also used for forward kinematics when sampling configurations. At execution time, a candidate is drawn from the map, MoveIt plans a collision-free motion to it, and the motion is executed in simulation. Section V covers which candidate is selected and in what order.

## IV. Base-Aware GNG Capability Map

Our core idea is a per-arm capability map. For each reachable location, the map stores a configuration that achieves it and a quality measure, rather than just a reachability bit. It is base-aware because the stored configuration includes the rail's degrees of freedom. We build one such map per arm, so the four arms across the two rails are described by four independently-queried maps (Section V).

### *A. Sampling Reachable Configurations*

We sample the 8-DOF configuration $\mathbf{q} = [d,\ \theta,\ q_1,\ \ldots,\ q_6]$ uniformly within the joint limits and compute the end-effector pose by forward kinematics with Pinocchio [20], recording the manipulability $w = \sqrt{\det(JJ^\top)}$ [11] for each sample. Because $d$ and $\theta$ are sampled alongside the arm joints, the resulting (pose, $\mathbf{q}$, manipulability) dataset already reflects the reach that base motion provides, not just a fixed arm's.

### *B. Growing the Map*

We fit a Growing Neural Gas network [12] to this dataset. Each node stores $[\mathbf{x} \mid \mathbf{q}]$, concatenating the end-effector position $\mathbf{x}$ (the task part) with the configuration $\mathbf{q}$ that produces it. The best-matching-unit search uses the task part alone, tiling nodes over the reachable workspace, while adaptation updates the whole vector, so each node's $\mathbf{q}$ converges to a representative configuration for its cell; edges form between co-activated nodes by competitive Hebbian learning. Every node is thus both a reachable location and a representative 8-DOF configuration (rail placement included), which is the sense in which the map is base-aware, since $\mathbf{q}$'s rail translation/rotation keeps different base placements of the same point in different nodes rather than averaged together. To stay faithful at the reachable boundary, where interior-only samples would settle inward, we pin a shell of nodes on the measured reach surface before growing the interior (Fig. 3).

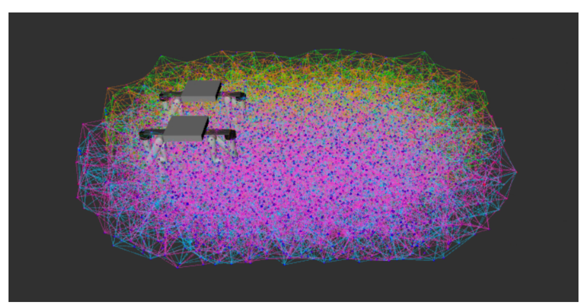


Fig. 3. The base-aware GNG capability map for one arm, rendered in RViz.

### *C. Capability Layers*

Beyond plain reachability, each node also carries a quality value, manipulability $w$ [11], low near the workspace edge and singular postures and high where the arm moves freely.

This is the layer separating our map from a binary reachability grid [9]; it is computed once offline with the same Pinocchio model [20] and available at query time at no extra cost.

## V. Energy-Aware Arm and Base Selection

Given the object pose from perception and the four capability maps, the selection stage decides which arm and which 8-DOF configuration to reach with, both together, via an energy cost evaluated over candidates drawn from the maps.

### A. The Energy Cost

We score a candidate configuration by a weighted sum of the effort of using it,

$$J = w_{gl}\left(\frac{\Delta_{lin}}{\rho_{gl}}\right) + w_{gr}\left(\frac{\Delta_{rot}}{\rho_{gr}}\right) + w_{arm}\left(\frac{\Delta_{arm}}{\rho_{arm}}\right) + w_{dist}\left(\frac{e}{\rho_{dist}}\right) - w_{manip}\left(\frac{m}{\rho_{manip}}\right), \quad (1)$$

where $\Delta_{lin}$, $\Delta_{rot}$, and $\Delta_{arm}$ are the rail-linear, rail-rotation, and summed arm-joint travel from the current state to the candidate, $e$ is the distance from the arm's current tool frame to the object, and $m$ is its manipulability [11]. The rail's linear and rotational axes carry separate weights because their units and their cost differ, metres of heavy-carriage travel against radians of rotation. Manipulability enters with a minus sign, so a more dexterous configuration is cheaper. Each term is divided by a fixed reference $\rho$ before its weight is applied, which makes the terms dimensionless and of comparable magnitude so that a weight reads directly as the priority of that term. We set each reference to the median value the term takes over a calibration set of picks, so that no term dominates the sum merely because of the units it is measured in. The weights themselves are fit by ordinary least squares, regressing the measured mechanical energy of executed trajectories on the six normalized terms over a calibration set of picks, rather than set by hand. Table I lists the calibrated weights and references.

TABLE I
Calibrated weights and normalization references for $J$.

| Term | Symbol | Weight | Reference $\rho$ |
|---|---|---|---|
| Rail linear travel | $\Delta_{lin}$ | 27.16 | 1.18 |
| Tool-to-object distance | $e$ | 12.78 | 1.45 |
| Rail rotation travel | $\Delta_{rot}$ | 3.08 | 1.24 |
| Manipulability (subtracted) | $m$ | 2.95 | 0.105 |
| Arm joint travel | $\Delta_{arm}$ | 0.09 | 7.18 |

The terms are proxies for the energy of the pick. Rail and arm travel are the mechanical work of moving each axis (the heavy rail dominates [6], [7]), and tool-to-object distance stands for the arm motion still needed to reach the object. A Coulomb/viscous friction model on the rail (mass- and $\mu$-derived, a stated assumption, not measured hardware) is what lets rail-linear travel dominate the fitted weights; a frictionless rail gave a far weaker fit ($R^2 = 0.055$) and put nearly all weight on arm travel instead. We do not claim $J$ is the exact pick energy, but Section VI shows minimizing it lowers the measured mechanical energy of the executed trajectory relative to the baselines [8].

### B. Pooling and Selection

For a target object, we pool the nodes within a map-spacing-scaled radius from each of the four arms' capability maps and score every candidate with $J$ on the same scale, so the pool always compares all four arms together, including arms on different rails. We sort the pool by $J$ and try candidates in order, interleaved across arms so no single arm consumes every attempt (the overall lowest-$J$ arm gets a small head start); for each candidate we solve IK to a top-down pre-grasp pose and ask MoveIt for a collision-free plan, plan-only (the arm never moves) until the first valid plan is selected (Fig. 4). With execution enabled, only the selected candidate is planned and executed, falling through to the next candidate if execution aborts on a scene change. The winning arm and its rail placement *are* the arm-and-base decision this paper sets out to make.

Fig. 4. The pipeline executing an energy-aware pick in Isaac Sim (`arm_2` approaching the banana).

## VI. Experiments

### A. Capability Map Characterization

Each arm's base-aware GNG capability map, built with 80,000 FK samples and a ceiling-height filter (2915 nodes: 600 pinned boundary, 2315 interior; $\lambda = 60$, 2 training epochs), has near-identical statistics across all four arms (median node spacing 0.090 m, mean 0.092 m, ~14,500 edges, 8 DOF per node), expected since gantry_2 is a $y$-mirror of gantry_1. Treating the rail as a DOF expands the reachable volume from 1.920 m$^3$ to 7.899 m$^3$ at 0.05 m voxel resolution (about $4\times$, as reported in Section I), and pinning a boundary shell before growing the interior cuts the reachable-edge shortfall from 0.221 m to 0.008 m.

### B. Selection-Policy Comparison

We compare four selection policies, energy, nearest, fixed, and random, over a 60-position grid of target poses spanning both rails. Each target is planned, but not executed, against the live MoveIt move-group node, with trajectory-energy computation enabled. The fixed baseline is run once per arm and reported as best/worst. Table II summarizes success rate and, over successful picks only, mean/median rail-linear travel, rail-rotation travel, summed arm-joint travel, measured mechanical energy (Pinocchio inverse dynamics), and plan time.

**Rail friction model.** The gantry rail's URDF joints originally carried no `<dynamics>` damping/friction, so this idealised energy could not see the cost of horizontal rail travel (both axes orthogonal to gravity), and $J$'s best single-term correlation with measured `traj_energy` topped out at $\rho \approx 0.31$. We add a modelling assumption, not measured hardware data. Coulomb friction $= \mu N$ ($N =$carried weight$\times g$, URDF link masses; $\mu = 0.15$, mid of a typical linear-guide range) plus a small viscous term on both rail joints (friction $=$ 27.6 N linear / 2.03 Nm rotational), applied explicitly since Pinocchio's `rnea` does not itself apply URDF friction (verified: `rnea` output was identical at zero vs. non-zero joint velocity without this fix). Re-fitting $w_*$/ref$_*$ by OLS on the five normalised $J$ terms (168 picks) reaches $R^2 = 0.857$, Spearman $= 0.909$ (up from $R^2 = 0.055/\rho \approx 0.31$), and gantry_lin becomes dominant ($w = 27.16$) while arm is nearly negligible ($w = 0.09$). This is a friction-dominated rail cost, recovering from the data alone the Section I premise that repositioning the base is the dominant cost. Table II reports the rerun under this model.

TABLE II
SELECTION-POLICY COMPARISON UNDER THE RAIL-FRICTION MODEL.

| Mode | Succ. | Rail lin. (m) | Rail rot. (rad) | Arm (rad) | Energy (J) |
|---|---|---|---|---|---|
| energy (ours) | 56/60 (93.3%) | 0.97/1.01 | 1.27/1.14 | 8.66/8.76 | **45.84**/52.34 |
| nearest | 56/60 (93.3%) | 1.19/1.41 | 1.37/1.33 | 7.94/7.50 | 48.28/54.79 |
| random | 56/60 (93.3%) | 1.17/1.28 | 1.36/1.35 | 8.10/7.69 | 47.31/51.72 |
| fixed (best: arm_4) | 39/60 (65.0%) | 1.14/1.21 | 1.22/1.23 | 7.47/7.17 | 49.79/55.78 |
| fixed (worst: arm_1) | 38/60 (63.3%) | 1.13/1.28 | 1.74/1.67 | 8.05/7.56 | 45.43/43.12 |

The four single-arm `fixed` baselines cluster at 63–65% success (arm_1 38/60 worst; arm_2/3/4 tied 39/60 best), well below the 93.3% every multi-arm policy reaches, so four-arm coverage is the dominant reachability gain (unchanged from the frictionless model). Fixed-arm energy figures are each over that arm's own smaller reachable set, so the meaningful energy comparison is `nearest` vs. `random`. Under friction, energy-aware selection has the lowest mean measured energy of the three multi-arm policies (45.84 J vs. nearest 48.28 J, $-5.1\%$, and random 47.31 J, $-3.1\%$; Fig. 5), a reversal from the frictionless model, where energy-mode's mean was *highest* (13.19 J vs. 12.01/12.87 J). This is a real but partial win. Median energy (52.34 J) beats nearest (54.79 J) but ties random (51.72 J), and a paired same-target comparison (56 rows) has energy beating nearest 57.1% and random 60.7% of the time, a real majority but not a sweep. Arm travel is now *higher* than nearest's (8.66 vs. 7.94 rad, opposite of frictionless) because the recalibrated weights de-prioritise arm travel once rail friction dominates, which is the intended friction-aware trade. In a representative case (target $x$=1.12, $y$= $-$ 0.60, $z$=1.15) energy chose arm_3 (0.64 m rail travel, 10.23 rad arm, 42.37 J) over nearest's arm_4 (1.15 m rail, 5.71 rad arm, 72.47 J), for 0.51 m less rail travel and 30.1 J lower energy. Fig. 4 shows this trade-off executing end to end.

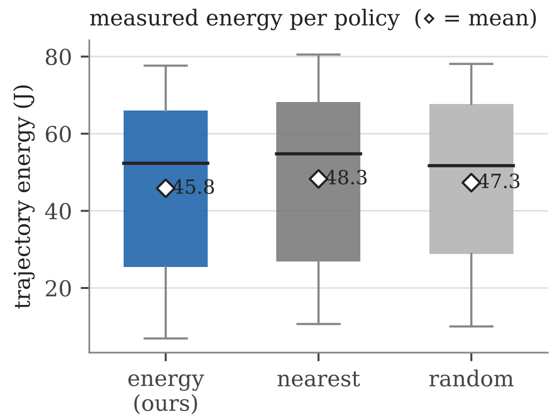



Fig. 5. Measured trajectory energy per selection policy (◇ = mean).

### C. End-to-End Pick and Reachability Classification

We validated the full perception-to-execution pipeline (Isaac ground-truth segmentation $\rightarrow$ localization $\rightarrow$ energy-aware arm/base selection $\rightarrow$ IK $\rightarrow$ planning $\rightarrow$ real Isaac-physics execution) on the 7 of 8 scene objects reachable-by-design (cracker_box, scissors, mustard_bottle, teddy_bear, banana, mug, bowl); the 8th, tomato_soup_can, is the unreachable-by-design negative example used below. With object poses fixed at asset-import time, we ran 7 objects $\times$ 5 trials $=$ 35 picks in round-robin order as a pipeline-integration demo, not a positional-sensitivity study (that role is filled by the selection-policy grid sweep of Section VI-B). Result: 35/35 (100%) pick success, zero failures in any category (detection, IK/unreachable, planning, execution). Mean time-to-pick was 39.6 s (median 28.1 s, range 10.5–184.4 s); scissors had the widest spread (110.4 s mean) from a transient settle mismatch that the executor's retry loop silently recovered by falling through to the next arm. Energy-mode distributed the 35 picks arm_2:10, arm_3:13, arm_4:12, arm_1:0; the zero for arm_1 is a property of this fixed layout, not the algorithm, which does pick arm_1 in 12/60 targets of the Section VI-B grid sweep. Perception used Isaac ground truth rather than YOLOE for these picks, following an earlier finding that YOLOE was unreliable on this simulator's synthetic imagery. A separate accuracy check with updated (`11m`) detector weights found this superseded. Matched against ground truth over 3864 detections, YOLOE reached 100% detection, 0.023 m mean localization error, and 0% false positives at conf$=0.25$ (rising to 15.2% false positives at conf$=0.10$), so YOLOE-driven end-to-end execution is now plausible future work rather than an expected failure.

**Reachable-vs-unreachable classification.** The same ground truth (7 objects picked 35/35 above, plus tomato_soup_can re-confirmed unreachable, with an IK error of $-31$ on all 4 arms) was compared against the GNG map's nearest-node-distance reachability test, swept over `reach_radius`$\in \{0.08, 0.12, 0.16\}$ m. Accuracy reached 100% (8/8) at `reach_radius`$\geq 0.12$ m (87.5% at 0.08 m), matching the independently-computed nearest-node distances exactly (0.120/0.152 m for the two boundary-closest objects) with zero false positives at any radius, the same signal `gantry_reach_executor` uses for IK candidates. A pre-existing diagnostic-only tool (`reachability_cloud` RViz coloring) additionally gates on an 8-node one-sided "enclosure" check not part of the real decision; with it active 2/8 objects read 0% at every radius (one-sided map fringe), and disabling it recovers the 100% result.

## VII. Conclusion

We presented an energy-aware selection cost $J$ that chooses both the arm and the full 8-DOF configuration, rail travel included, on a ceiling-mounted dual-gantry quad-arm robot, built on a base-aware GNG capability map that treats the rail as a first-class degree of freedom, with boundary-seeding keeping the map accurate to within centimetres of the true reach surface. In simulation, treating the rail as a DOF gives a $\sim 4\times$ reachable-workspace gain over an arm-only baseline, and under a rail-friction model energy-aware selection reaches the lowest mean measured mechanical energy of the three multi-arm policies compared, a real but partial win, reported with its median and pairwise caveats (Section VI-B). The full perception-to-execution pipeline executed 35/35 end-to-end picks and classified reachable-vs-unreachable objects at 100% accuracy from the same capability map. All results come from Isaac Sim; validating on the physical ceiling-rail platform, and studying the sim-to-real gap in the calibrated weights of $J$, is left for future work.

## References


[1] United Nations, Department of Economic and Social Affairs, Population Division, "World Population Prospects 2024: Summary of Results," United Nations, New York, NY, USA, 2024.

[2] M. Lisondra, B. Benhabib, and G. Nejat, "Embodied AI with foundation models for mobile service robots: a systematic review," *Robotics*, vol. 15, no. 3, p. 55, Mar. 2026, doi: 10.3390/robotics15030055.

[3] C. C. Kemp, A. Edsinger, and E. Torres-Jara, "Challenges for robot manipulation in human environments [Grand Challenges of Robotics]," *IEEE Robotics & Automation Magazine*, vol. 14, no. 1, pp. 20–29, 2007, doi: 10.1109/MRA.2007.339604.

[4] I. Lalonde, J. Denis, M. Lamy, and A. Girard, "A Dual-Motor Actuator for Ceiling Lift with High-Force and High-Speed Capabilities," *Actuators*, vol. 14, no. 2, p. 92, Feb. 2025, doi: 10.3390/act14020092.

[5] Y. P. D. Kaluarachchi, T. D. S. S. Senarathna, A. V. P. Lakshan, H. A. G. C. Premachandra, Y. W. R. Amarasinghe, and W. A. D. M. Jayathilaka, "A Novel Hybrid Robot Configuration for Enhanced Accessibility and Space Efficiency," in *Selected Proceedings of the 2nd International Engineering Research Symposium (IERS 2024)*, Springer, 2025, doi: 10.1007/978-981-96-1399-1_22.

[6] B. Shao, N. Cao, Y. Ding, X. Wang, F. Gu, and C. Chen, "MoMa-Pos: an efficient object-kinematic-aware base placement optimization framework for mobile manipulation," arXiv:2403.19940, 2024.

[7] L. Naik, S. Kalkan, S. L. Sørensen, M. B. Kjærgaard, and N. Krüger, "BaSeNet: a learning-based mobile manipulator base pose sequence planning for pickup tasks," in *Proc. IEEE/RSJ Int. Conf. Intelligent Robots and Systems (IROS)*, 2024, pp. 3666–3673, doi: 10.1109/IROS58592.2024.10802615.

[8] A. Gupta, "Energy efficiency in robotics software: a systematic literature review (2020–2024)," arXiv:2508.12170, 2025.

[9] F. Zacharias, C. Borst, and G. Hirzinger, "Capturing robot workspace structure: representing robot capabilities," in *Proc. IEEE/RSJ Int. Conf. Intelligent Robots and Systems (IROS)*, 2007, pp. 3229–3236, doi: 10.1109/IROS.2007.4399105.

[10] A. Makhal and A. K. Goins, "Reuleaux: robot base placement by reachability analysis," in *Proc. IEEE Int. Conf. Robotic Computing (IRC)*, 2018, pp. 137–142, doi: 10.1109/IRC.2018.00028.

[11] T. Yoshikawa, "Manipulability of robotic mechanisms," *Int. J. Robotics Research*, vol. 4, no. 2, pp. 3–9, 1985.

[12] B. Fritzke, "A growing neural gas network learns topologies," in *Advances in Neural Information Processing Systems 7 (NIPS)*, MIT Press, 1995, pp. 625–632.

[13] A. Wang, L. Liu, H. Chen, Z. Lin, J. Han, and G. Ding, "YOLOE: real-time seeing anything," in *Proc. IEEE/CVF Int. Conf. Computer Vision (ICCV)*, 2025, pp. 24591–24602, doi: 10.1109/ICCV51701.2025.02280.

[14] S. Gao, M. Pagnucco, T. Bednarz, and Y. Song, "NVIDIA Isaac Sim: enabling scalable, GPU-accelerated simulation for robotics," arXiv:2606.03551, 2026.

[15] N. Kubota, "Multiscopic Topological Twin in Trailer Living Laboratory: Plenary Talk," in *Proc. IEEE 22nd Int. Symp. Computational Intelligence and Informatics and 8th IEEE Int. Conf. Recent Achievements in Mechatronics, Automation, Computer Science and Robotics (CINTI-MACRo)*, 2022, pp. 11–12, doi: 10.1109/CINTI-MACRo57952.2022.10029665.

[16] S. Duan, P. Ren, N. Jiang, Z. Che, J. Tang, Z. Fan, Y. Sun, and W. Wu, "RoboPARA: Dual-Arm Robot Planning with Parallel Allocation and Recomposition Across Tasks," in *Proc. Int. Conf. Learning Representations (ICLR)*, 2026. [Online]. Available: https://openreview.net/forum?id=b9GH5mMBfG

[17] Kinova Robotics, "Gen3 lite robot – user guide / product specifications," [Online]. Available: https://www.kinovarobotics.com

[18] D. Coleman, I. A. Şucan, S. Chitta, and N. Correll, "Reducing the barrier to entry of complex robotic software: a MoveIt! case study," *Journal of Software Engineering for Robotics*, vol. 5, no. 1, pp. 3–16, May 2014, doi: 10.6092/JOSER_2014_05_01_p3.

[19] I. A. Şucan, M. Moll, and L. E. Kavraki, "The Open Motion Planning Library," *IEEE Robotics & Automation Magazine*, vol. 19, no. 4, pp. 72–82, Dec. 2012, doi: 10.1109/MRA.2012.2205651.

[20] J. Carpentier, G. Saurel, G. Buondonno, J. Mirabel, F. Lamiraux, O. Stasse, and N. Mansard, "The Pinocchio C++ library: a fast and flexible implementation of rigid body dynamics algorithms and their analytical derivatives," in *Proc. IEEE/SICE Int. Symp. System Integration (SII)*, 2019, pp. 614–619, doi: 10.1109/SII.2019.8700380.